Binary block systems and transitive subgroups

Human Compression project: a component of the L044 recovery

Write $G_I = \langle \rho_j : j \notin I \rangle$ for a group with distinguished generators $\rho_0, \ldots, \rho_{r-1}$, and $G_i = G_{\{i\}}$. A binary block system is a partition into two nonempty blocks which the group permutes. The following observation uses the actual partitions, including the action of a generator omitted from one of their stabilizers.

Lemma 1. *Suppose $G \leq S_\Omega$ is primitive, where $|\Omega| = n > 2$. Let I have t members, suppose G_I is transitive, and suppose each G_i, for $i \in I$, preserves a binary block system. Then their common refinement has exactly 2^t equally sized blocks. The action of G_I on them is the regular translation group $\mathbb{F}_2^t$. In particular 2^t divides n. If $n = 2^t$, then G_I itself is this elementary abelian group in its regular action.*

Proof. Choose a function $f_i : \Omega \to \mathbb{F}_2$ specifying the two blocks of the ith partition. For $g \in G_i$, there is a constant $\chi_i(g)$ such that

$$f_i(gx) = f_i(x) + \chi_i(g) \quad (x \in \Omega).$$

The restriction of each χ_i to G_I is a homomorphism. Let $F = (f_i)_{i \in I}$, and let V be the image of the joint homomorphism $\chi : G_I \to \mathbb{F}_2^t$. Transitivity gives $F(\Omega) = F(x_0) + V$. It also makes all nonempty fibres of F equally sized.

If V were proper, some nonzero vector $(a_i)_{i \in I}$ would annihilate it. Thus $\sum_i a_i f_i$ would be constant on Ω. Choose s with $a_s = 1$. For every $i \neq s$ in I, the generator ρ_s belongs to G_i, so $f_i(\rho_s x) - f_i(x)$ is constant. The displayed linear relation then makes $f_s(\rho_s x) - f_s(x)$ constant as well. Hence ρ_s preserves the sth partition. All other distinguished generators already preserve it because they belong to G_s. This contradicts primitivity of G. Therefore $V = \mathbb{F}_2^t$, proving the refinement and divisibility assertions. If each fibre is a singleton, the actual action is the faithful regular translation action, proving the last assertion. □

Lemma 2. *Let a transitive group be generated by a finite distinguished family, and let B be generated by b members of that family. If B has a point orbits, some transitive subgroup is generated by those b members and at most $a - 1$ additional members.*

Proof. As long as the current subgroup is intransitive, some distinguished generator joins two of its orbits; otherwise every generator would preserve every current orbit. Adjoining that generator reduces the number of orbits by at least one. At most $a - 1$ repetitions suffice. No action of the newly adjoined generator on the old orbit set is assumed. □

Theorem 3. *Let $G = A_n$ in its natural action, $n \geq 16$, with a string C-marking of rank $r \geq n/2$. Suppose that every transitive G_i preserves a binary block system. If some $G_{\{i,j\}}$ is transitive, then $n = 16$ and $r = 8$.*

In that remaining case there is a set J of five distinguished labels such that $T = \langle \rho_j : j \in J \rangle$ is transitive. After a change of coordinates its points are $(a, b) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, and

$$T \leq W = \left\{ (a, b) \longmapsto (a + u, b + v(a)) : u \in \mathbb{F}_2^3,\ \sum_a v(a) = 0 \right\}. \tag{1}$$

In particular $|W| = 1024$. The induced marking on T has the full intersection property, with its original labels.

Proof. Put $H = G_{\{i,j\}}$. By Lemma 1, its two binary partitions refine to four blocks of size $l = n/4$, and its image on them is $\mathbb{F}_2^2$. Choose two distinguished generators x, y of H whose images form a basis, and put $B = \langle x, y \rangle$. Each B-orbit meets all four blocks equally, so there are at most l such orbits.

If x, y commute, B is a Klein four-group with faithful regular action on the four blocks. All its point orbits have size four, and it has exactly l orbits. Lemma 2, applied inside H, gives a transitive subgroup on at most $l + 1$ distinguished generators. Its complement has at least $r - l - 1 \geq l - 1$ labels. Every maximal omission at one of those labels is transitive, since it contains that subgroup. Lemma 1 implies

$$2^{l-1} \mid 4l. \tag{2}$$

For $l \geq 6$, $2^{l-1} > 4l$: this holds at six and the ratio increases thereafter. For $l = 5$, sixteen does not divide twenty. Thus $l = 4$ is the only possibility with $n \geq 16$.

If x, y do not commute, at least one B-orbit has more than four points. Indeed on a four-point orbit the two generators act as their commuting translations on the four blocks. Hence B has at most $l-1$ orbits. Lemma 2 now supplies a transitive subgroup on at most l distinguished generators, containing B. Its omitted label set has size at least l. For $l \geq 5$ this contradicts $2^l > 4l$ and Lemma 1. For $l = 4$, that omitted set must have exactly four members; the last assertion of Lemma 1 makes the transitive subgroup abelian, contradicting the presence of x, y. This excludes the noncommuting case altogether.

We therefore have $n = 16$ and a transitive subgroup on a set J_0 of at most five distinguished labels. Put $t = r - |J_0|$. Lemma 1 gives $t \leq 4$, so $r \leq 9$. If $r = 9$, necessarily $|J_0| = 5$ and $t = 4$. The subgroup is then elementary abelian of order sixteen. The intersection property makes its five distinguished generators independent, impossible in a vector space of dimension four. Thus $r = 8$. Enlarge J_0, if necessary, to a set J of five labels; its generated subgroup remains transitive.

For the three omitted labels, Lemma 1 gives eight pairs, indexed by their three binary coordinates a. The subgroup T acts on these pairs by translations $a \mapsto a + u$. Choose the two point names b in each pair arbitrarily. Every permutation inducing a translation has the form in (1). A translation on the eight pairs is even on the sixteen points: each exchanged pair of pairs contributes two transpositions. The remaining sign is $(-1)^{\sum_a v(a)}$. Since $T \leq A_{16}$, the stated even-sum condition holds. There are eight translations and 2^7 permitted functions v, proving the order assertion. Intersection properties on subsets of J are inherited from the original C-marking. □

The remaining finite obligation. Degree sixteen still requires completions of transitive five-mark subgroups in W. The $\binom{8}{5} = 56$ label sets J give twenty-eight cases up to reversal. Identify the omitted labels with the three coordinates of a. Each omitted generator preserves the other two binary partitions, so belongs to the even part of $(S_4)^4 \rtimes \mathbb{F}_2^2$ on their four cells. It must break its own partition. String relations, generation and all subset intersections remain obligations; no exhaustion is asserted here.

Attribution and dependencies. The target is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 3.4. The source uses intersecting binary partitions and a four-partition divisibility argument. Lemma 1 supplies their general independence proof; orbit growth replaces the separate large-degree and degree-twenty/twenty-four arguments. Other imprimitive branches must supply our binary-system hypothesis. The case where every double omission is intransitive remains separate. No priority for binary partitions, finite computation, conjecture, Lean verification or whole-original credit is asserted.